Supplementary Information on "Mining higher-order triadic interactions"

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SUPPLEMENTARY INFORMATION ON THE TRIM ALGORITHM

Estimation of the Mutual Information

The Mutual Information $MIz(m)$ between variables X and Y conditioned on the value z_m of the variable Z (putative regulator node of the triadic interaction) is calculated using k-nearest-neighbour entropy estimation to be able to estimate it only by using the time series values. [1–3], $MIz(m)$ is defined as:

$$MIz(m) = \int dx \int dy \mu(x, y|z_m) \log \left(\frac{\mu(x, y|z_m)}{\mu(x|z_m)\mu(y|z_m)} \right). \quad (1)$$

By assuming some metric, neighbours of a chosen point are ranked by distance. The local density of points is estimated by considering the average distance of the k nearest neighbours. This local density is used to estimate the logarithms of the entropies using the Kozachenko-Leonenko-Estimator [1].

Pipeline of TRIM

In order to select for the significant triples, we combine information coming from the two considered null models (the surrogate randomised data and the maximum likelihood Gaussian model) and the entropic score S. First we select the set of triples of interest, and we choose which observable to consider, either Σ or T. Note that we have found that the results obtained considering either one of the two observables are highly correlated (see section on gene-expression results).

With respect to the randomized null model, the Gaussian null model more efficiently captures correlations between the dynamical state of the three considered nodes due to network effects. However, the second null model is more sensitive to the presence of outliers in the data but it is still focusing only on the three nodes in the triple and neglects more collective network effects. In order to further reduce the set of relevant triples, we screen out triples with low entropic score. Thus, we define the following TRIM pipeline:

(a) For each link of interest between variables X and Y consider all the triples X,Y, Z where Z is any given possible regulator node (any node different from X and Y) and calculate Θ_Σ or Θ_T together with their corresponding p-values. Select only triples with p-values smaller than a threshold (typically taken 5×10^{-3} or 1×10^{-3}).

(b) Screen out all the triples which have a high p-value according to the second (Gaussian) null model. Here a high p-value indicates a p-value higher than a threshold typically taken the same as in point (a)).

(c) Rank the remaining triples that rank high according to the Θ score calculated according to the randomized surrogate data. For each link between node X and Y select only the top 5 triples [X,Y,Z].

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(d) From the remaining triples screen out all the triples which have a low entropic score S (for instance $S < 0.5$).

SUPPLEMENTARY INFORMATION ON THE TRIADIC PERCEPTRON MODEL (TPM)

Derivation of Eq.(6) of the main text

Let us consider the stochastic dynamics Eq.(5) in the main text, driving the TPM in the absence of triadic interactions, i.e.

$$\frac{d\mathbf{X}}{dt} = -(\mathbf{L} + \alpha\mathbf{I})\mathbf{X} + \Gamma\boldsymbol{\eta}(t) \tag{2}$$

where $\mathbf{L}$ is the graph Laplacian of the networks $\alpha, \Gamma \in \mathbb{R}^+$ are two real parameters of the model, and $\boldsymbol{\eta}$ is a Gaussian white noise with

$$\langle \eta_i(t) \rangle = 0, \quad \langle \eta_i(t)\eta_j(t') \rangle = \delta_{ij}\delta(t - t'). \tag{3}$$

Here we want to derive Eq.(6) of the main text expressing the correlation matrix $C_{ij} = \mathbb{E}((X_i - \mathbb{E}(X_i))(X_j - \mathbb{E}(X_j)))$ at equilibrium, i.e.

$$C_{ij} = \frac{\Gamma^2}{2}[\mathbf{L} + \alpha\mathbf{I}]^{-1}_{ij}. \tag{4}$$

The derivation of this results is a straightforward outcome of the solution of the Fokker-Planck equation associated to the stochastic dynamics (2). Indeed the Fokker-Planck equation for the probability density function $P(\mathbf{X})$ of observing $\mathbf{X}$ reads

$$\frac{\partial P(\mathbf{X})}{\partial t} = -\sum_{i=1}^{N} \frac{\partial}{\partial X_i}[(\mathbf{L} + \alpha\mathbf{I})\mathbf{X}]_i P(\mathbf{X}) + \frac{\Gamma^2}{2}\sum_{i,j=1}^{N} \frac{\partial^2}{\partial^2 X_i} P(\mathbf{X}). \tag{5}$$

At stationarity, by imposing $\partial P(\mathbf{X})/\partial t = 0$ and solving for $P(\mathbf{X})$, we obtain a multivariate Gaussian distribution

$$P(\mathbf{X}) = \mathcal{C} \exp\left[-\frac{1}{2}\mathbf{X}^\top \mathbf{C}^{-1}\mathbf{X}\right], \tag{6}$$

where $\mathcal{C}$ is the normalization constant, and $\mathbf{C}$ indicates the covariance matrix is given by Eq.(4).

SUPPLEMENTARY INFORMATION ON THE TPM DYNAMICAL BEHAVIOUR

Here we provide supplementary information on the dynamics of the TPM (Eq. 2) on the network with $N = 10$ nodes, $L = 12$ edges and $\hat{L} = 5$ regulatory interactions shown in Figure 4 of the main text. We consider the time series obtained by integrating the stochastic dynamics of the proposed dynamical model for triadic interactions within continuous variables. The time series is simulated up to a maximum time $t_{\max} = 4000$ with a $dt = 10^{-2}$ leading to 4×10^5 data points. For our analysis we consider the last $200,000$ data points sampled every fifth point leading to $40,000$ time steps. In the Supplementary Figures S1 and S2 we report the analysis conducted on the triadic triple [4,9,5] already considered in Figure 4 but for different parameter values: for Figure S1 we have $\alpha = 0.01, \Gamma = 10^{-2}$, for Figure S2 we have $\alpha = 0.05, \Gamma = 5 \times 10^{-2}$. The other parameter values are: $\hat{T} = 10^{-3}, w^+ = 8, w^- = 0.5$, number of bins $P = 400$.

SUPPLEMENTARY INFORMATION ON ANALYSIS OF GENE-EXPRESSION DATASET

Methods

We choose the gene-expression associated with Acute Myeloid Leukemia (AML) extracted from the Grand Gene Regulatory Network Database [4, 5] for our TRIM analysis. Testing all possible combinations of triples of genes

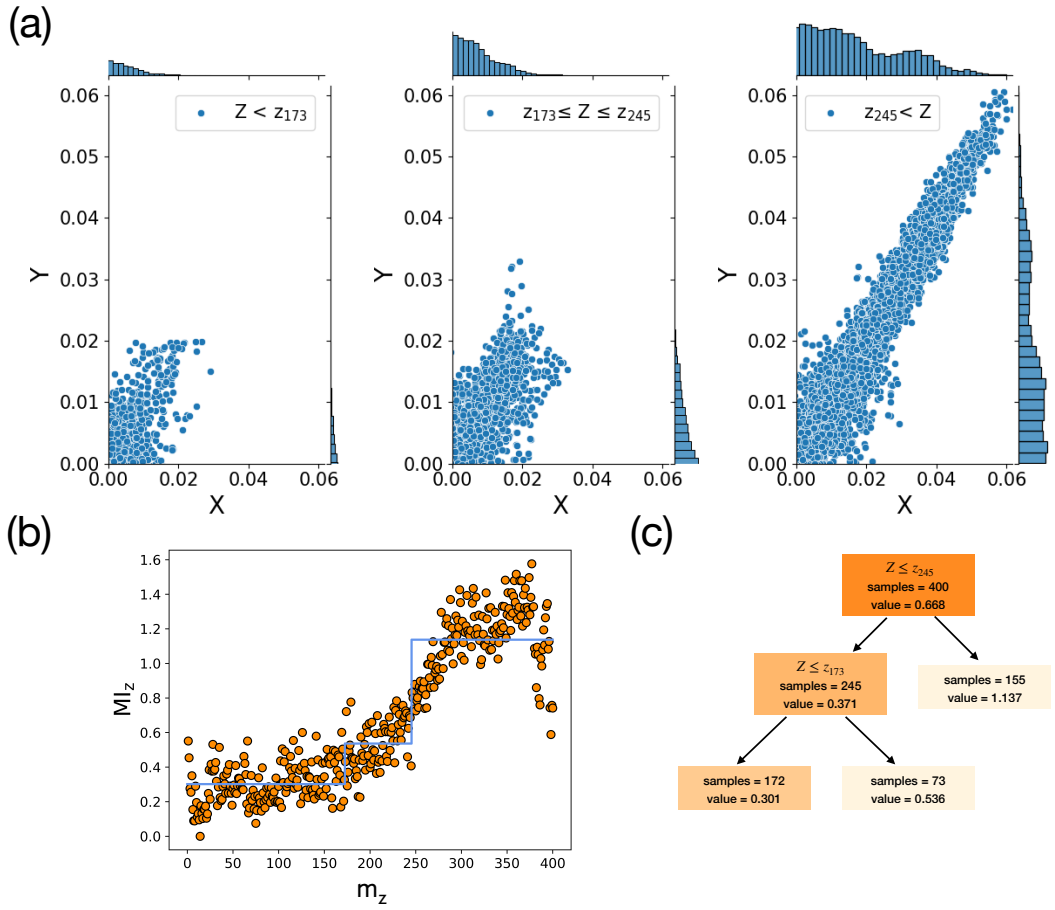


FIG. S1: Exemplary results obtained for a triple of nodes involved in a triadic interaction in the continuous model with triadic interactions. The joint distributions of variables X and Y conditional on the values of Z is shown in panel (a). Panel (b) displays the functional behaviour of MIz as a function of the values of z_m which clearly departs from the constant behaviour expected in absence of triadic interactions. Panel (c) presents the decision tree for fitting the MIz functional behaviour and determining the range of values of Z for which the most significant differences among the joint distributions of the variables X and Y conditional to Z are observed. The time series is simulated up to maximum time $t_{\max} = 4000$ with $dt = 10^{-2}$. For the analysis we consider $40,000$ time steps. The parameters of the model are: $\alpha = 0.01, \hat{T} = 10^{-3}$, $\Gamma = 10^{-2}$, $w^+ = 8, w^- = 0.5$, number of bins $P = 400$. The analysis is done for the triple [4,9,5], of the network in Figure 3 of the main text, which is triadic.

is computationally too demanding. To still make the analysis rigorous, we focus on the most important genes and the genes we suspect are most likely to be involved in triadic interactions. Hence, we utilize the Protein-Protein Interaction (PPI) network associated with AML, which captures digenic processes, to choose edges forming the triples in our study. This selection is motivated by studies indicating that for gene expression data, most of the genes involved in trigenic processes are also involved in digenic processes [6]. We consider all the genes in the AML datasets that are connected in the PPI network. Each edge in the PPI network has an assigned edge weight. This allows us to calculate the maximum spanning tree (MST) (Figure 7). The MST has diameter 16 and includes 622 nodes and 621 edges. 'EP300' is the node with the largest node degree. To further reduce the number of triples that are being analysed, only triples where the distance of potential regulator node and edge are at least 4 in the maximal spanning tree are being chosen. This is to reduce the effect of the triple being a triangle where all nodes are connected by an edge and to filter out dependencies from nodes of the triple being connected trough only a small number of edges.

Using this strategy $343,194$ triples are being analysed. To further decrease computational time, the number of realisations that the Σ or T values are being compared to is adjusted based on the confidence interval for a triadic interaction of the triple being investigated. First 10 realisations are run. If the p_Σ-value is bigger than 0.3, the triple is being categorised as non-triadic. In the other case, 100 realisations will be run and if the p_Σ-value of the new result is smaller than 0.05, 1000 realisations will be run. This is to quickly exclude any triplets that have extreme low probability of being triadic and only focus on the triples where there is uncertainty whether they are triadic or not.

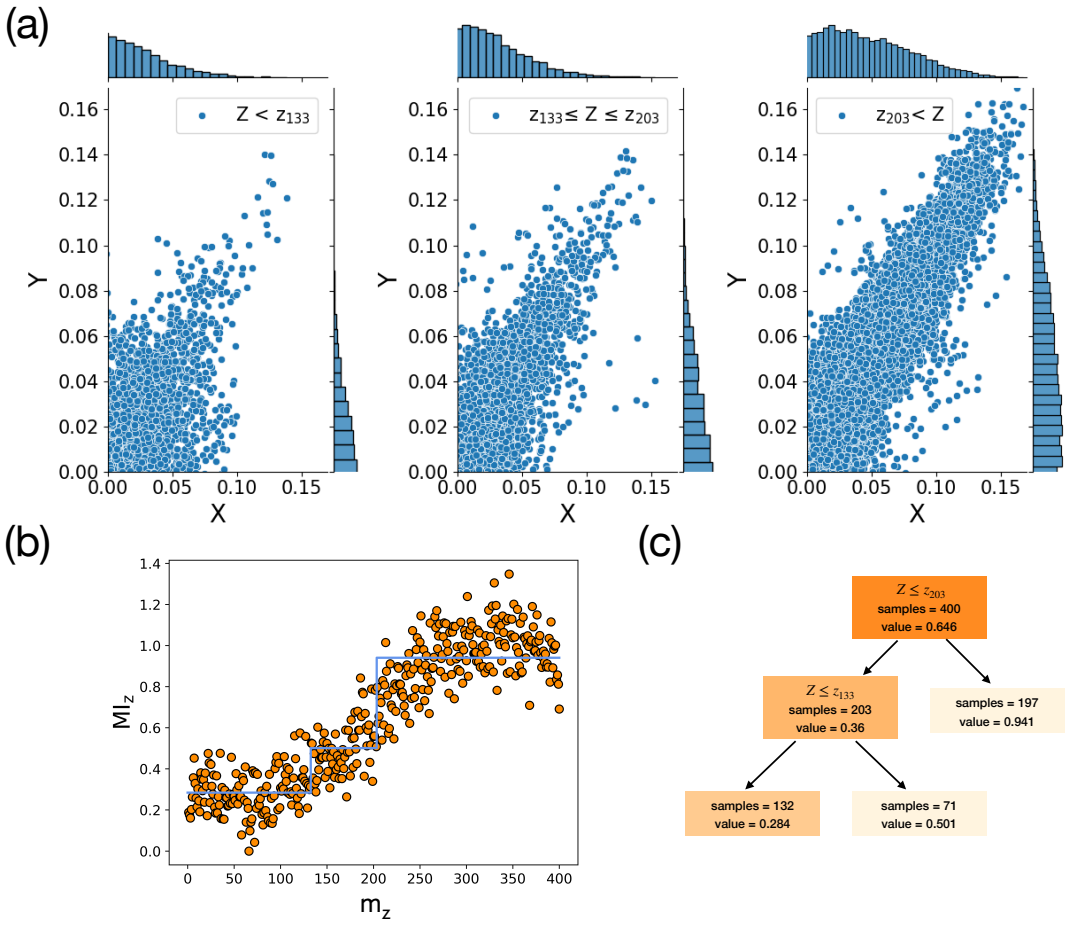


FIG. S2: Exemplary results obtained for a triple of nodes involved in a triadic interaction in the continuous model with triadic interactions. The joint distributions of variables X and Y conditional on the values of Z is shown in panel (a). Panel (b) displays the functional behaviour of MIz as a function of the values of z_m which clearly departs from the constant behaviour expected in absence of triadic interactions. Panel (c) presents the decision tree for fitting the MIz functional behaviour and determining the range of values of Z for which the most significant differences among the joint distributions of the variables X and Y conditional to Z are observed. The time series is simulated up to maximum time $t_{\max} = 4000$ with $dt = 10^{-2}$. For the analysis we consider $40,000$ time steps. The parameters of the model are: $\alpha = 0.05, \hat{T} = 10^{-3}, \Gamma = 5 \times 10^{-2}, w^{+} = 8, w^{-} = 0.5$, number of bins $P = 400$.The analysis is done for the triple [4,9,5], of the network in Figure 3 of the main text, which is triadic.

Our analysis of the data showed that several genes had an expression profile dominated by outliers that can strongly affect our analysis (see for an example the gene-expression profile of gene *TGIF2LX* in Figure S3). For each gene i we define an outlier score O_i given by

$$O_i = \max_{\alpha} \frac{|e_{i\alpha} - \langle e_i \rangle|}{\sqrt{\langle e_i^2 \rangle - \langle e_i \rangle^2}} \tag{7}$$

where $e_{i\alpha}$ is the gene-expression of gene i in sample/patient α, $\langle e_i^k \rangle = \sum_\alpha e_{i\alpha}^k / Q$ where Q is the number of samples. The maximum z-score O_i defined in Eq.(7) is a good measure to detect outliers. We thus remove the top 50 genes that score highest according to the O-score, corresponding to a cut-off $O \simeq 6$ as can be retrieved from a Figure S4. These are the outliers for which triples containing them are removed: [*TGIF2LX*, *DBX2*, *GSX1*, *POU4F2*, *EVX2*, *EN2*, *ONECUT1*, *PAX2*, *POU1F1*, *MNX1*, *HOXC10*, *BARX2*, *HOXD12*, *HOXC12*, *NKX2-5*, *FOXC2*, *HOXC11*, *PROX1*, *HMX2*, *FOXF1*, *HOXC8*, *NR0B1*, *PAX1*, *GSC*, *DMRT3*, *HOXD13*, *POU3F3*, *FOXA2*, *HOXD9*, *HNF4G*, *FOXF2*, *SP8*, *HMX3*, *TLX1*, *FOXD1*, *FOXA1*, *PHOX2B*, *SOX21*, *DMRT1*, *FOXQ1*, *VSX1*, *HOXC6*, *EGR4*, *NKX2-1*, *ZIC2*, *LBX1*, *RXRG*, *EMX2*, *SPZ1*]. Having removed the outlier genes from our analysis, we test for triadic interactions using the randomisation null model while the Gaussian null model, being more sensitive to the outliers is only used to exclude triples that are not significant.

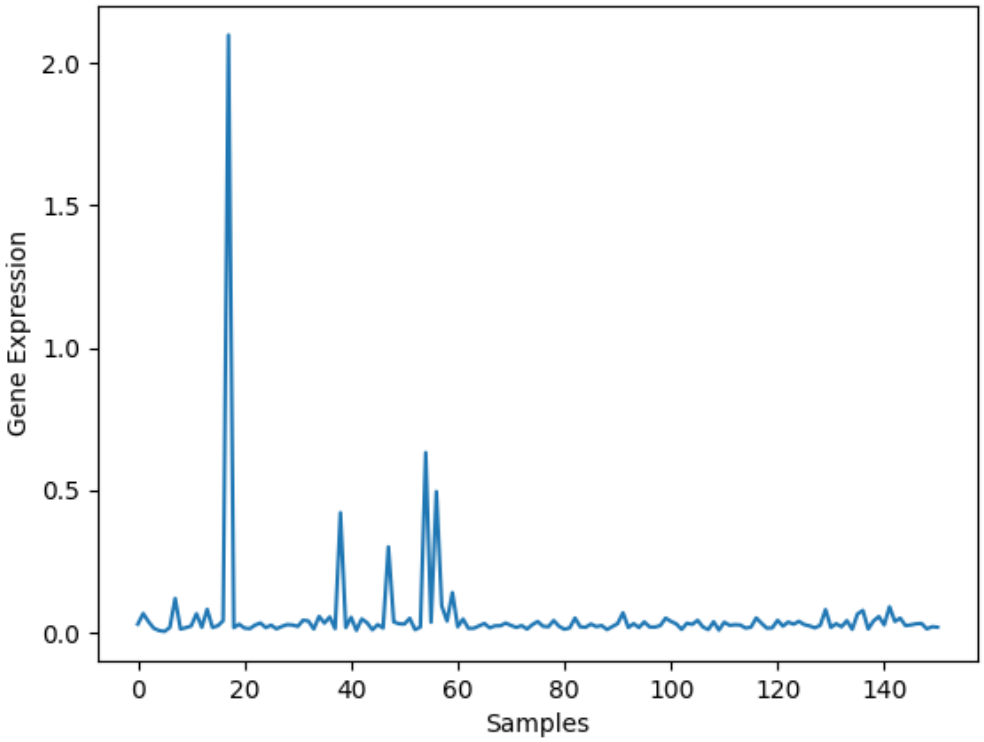


FIG. S3: Samples of gene *TGIF2LX*: Strength of the expression of the gene plotted against each individual patient tested. The biggest outlier is at position 17 with a value of 2.097, whereas the remaining values are smaller than 0.65.

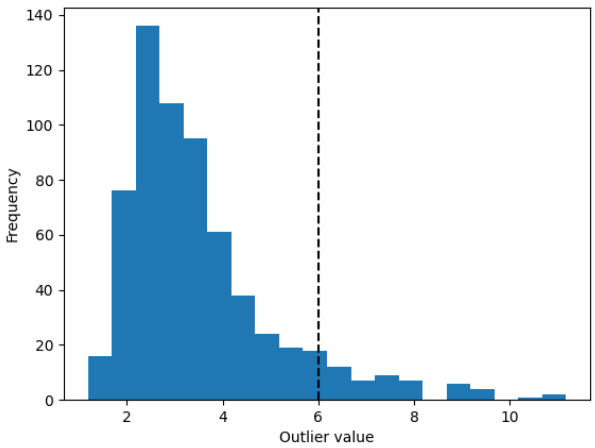


FIG. S4: Cut-off for the top 50 genes with high outlier values. The black vertical line indicates the cut-off.

For each edge, we select only the five triples with the highest Θ_Σ-value because these will be the most likely candidates for a triadic interaction. This is consistent with what we find for the synthetic data where triples containing the same edge can have higher Θ_Σ or p_Σ values but only the highest ranking is the true triadic triple. To verify that these are triadic and the dependencies do not stem from edges connecting the regulatory node with the other two nodes in the triple, we repeat the analysis for a subset of these triples with the Gaussian version of the null model

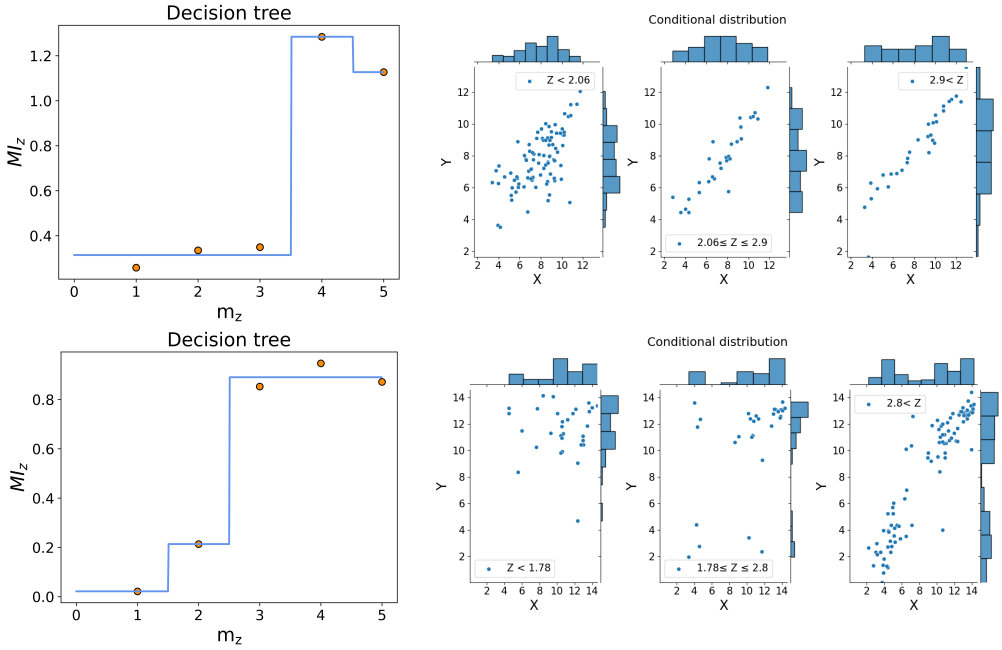


FIG. S5: Decision trees belonging to the conditional distribution of the triples shown in Figure 8. Entropy for the top triple is $S = 0.64$, entropy for the bottom triple is $S = 0.60$.

TABLE S1: Comparison of the two measures Θ_Σ and Θ_T using Kendall-Tau analysis for the Top 50 Triples in our Gene Expression Analysis.

	statistic	p-value
Gaussian: $\Theta_\Sigma - \Theta_T$	0.864	8.12×10^{-19}
Random: $\Theta_\Sigma - \Theta_T$	0.703	5.93×10^{-13}

realisations.

Supplementary results

The analysis reported in Figure 8 of the main-body of the paper focuses on the results obtained by using the Σ statistics. As supplementary material, in Figure *S5* we report the decision trees corresponding to the conditional distribution of the triples shown in Figure 8 and in Figure *S6* we report the triples screened out according to the TRIM pipeline because they have a p-value greater than 0.001 according to the Gaussian null model.

In Table *S1* we use Kendall's Tau to compare the ranking obtained using the two Θ_Σ and Θ_T. The Kendall's Tau is higher than 0.7 for both random and Gaussian null models so those measures are very similar.

As mentioned in the main text the TRIM algorithm can be potentially used to detect triadic interactions with non-monotonic behaviour. The joint conditional distributions and the decision tree of one exemplary non-monotonic triples is shown in Figure *S7*.

The TRIM algorithm was also used to mine triadic interactions among all the triples containing genes that have been found to be highly relevant for triadic interactions from a biological point of view. These genes are *HOXA1*, *HOXA2*, *HOXA3*, *HOXA4*, *HOXA5*, *HOXA6*, *HOXA7*, *HOXA8*, *HOXA9*, *HOXB1*, *HOXB2*, *HOXB3*, *HOXB4*, *HOXB5*, *HOXB6*, *HOXB7*, *HOXB8*, *HOXB9*, *PBX3*, *MEIS1* [7], [8], [9]. There are 125,244 triples of this type, 260 have been found to be candidates after applying the TRIM pipeline. Especially we observe that the method is able to give a high significance score to the triple *HOXA9*, *HOXA3*, *MEIS1* that has p_Σ-value 0.001 and Θ_Σ greater than 6 which is a triple of overexpressed genes in AML [7]. This triple together with the highest ranking biological triples by Θ_Σ are shown in Table *S2*.

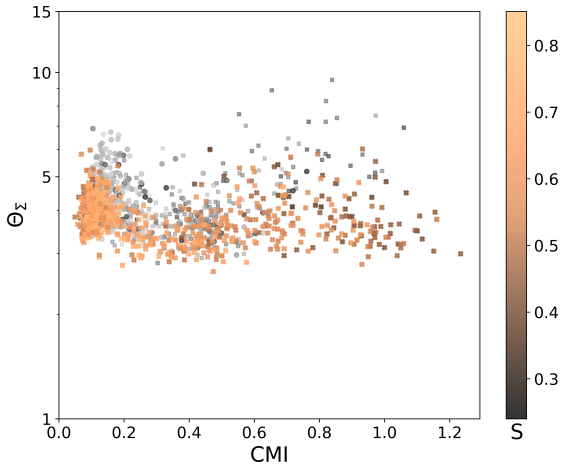


FIG. S6: The role of the Gaussian null model in screening out triples in the TRIM pipeline applied to the AML gene expression data. The scatter plot displays Θ_Σ (Y-axis) verse the CMI (X-axis). The colour corresponds to the value of our entropic score S. Here only triples that have p-value 0.001 or less in the randomization null model are shown. Crosses (in colour) are triples that have p-value higher than 0.001 in the Gaussian null model and therefore they are disregarded from the analysis according to the TRIM pipeline. Symbols in gray include the triples that are retained and that figure in Figure 8 of the main body of the paper. Specifically these include triples whose links appear in the MST (circles), and triples involving genes with biological relevance (squares).

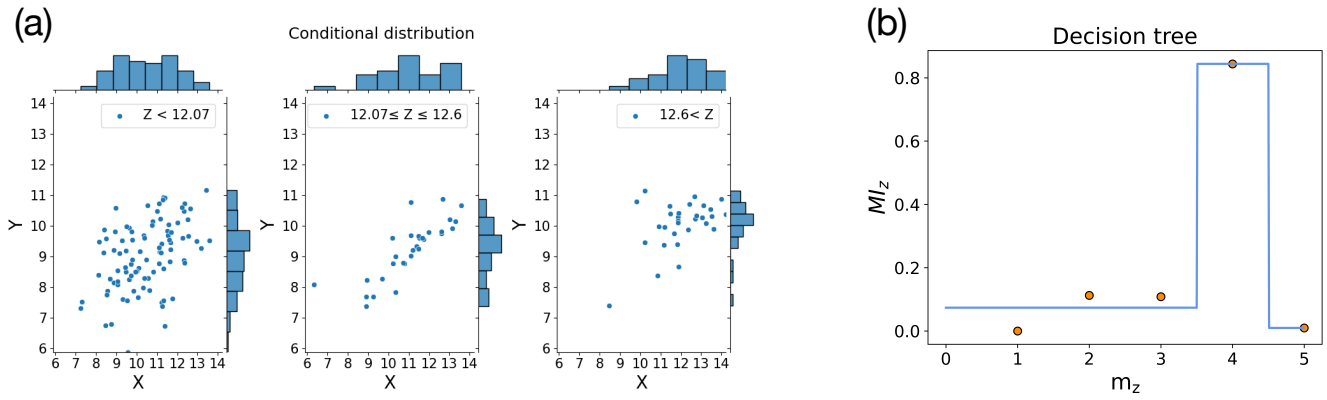


FIG. S7: An example of non-monotonic triadic interaction. The figure (a) display the conditional joined distributions for the triple X = *BCL6*, Y = *PPARD*, Z = *IRF5*, $\Theta_\Sigma = 4.23$, $p_\Sigma = 0.001$, $\Sigma = 0.32$, $S = 0.71$, and (b) the decision tree of the same triple.

Moreover in Table *S3* we report the data for the top 50 candidates of triadic interactions obtained using our procedure.

Finally in Table *S4* we provide the list of the literature on the genes involved on the top scoring triples detected by our analysis of the AML gene-expression data.

[1] Lyudmyla F Kozachenko and Nikolai N Leonenko. Sample estimate of the entropy of a random vector. *Problemy Peredachi Informatsii*, 23(2):9–16, 1987.

[2] Alexander Kraskov, Harald Stögbauer, and Peter Grassberger. Estimating mutual information. *Physical review E*, 69(6):066138, 2004.

[3] Brian C Ross. Mutual information between discrete and continuous data sets. *PloS one*, 9(2):e87357, 2014.

TABLE S2: Highest ranking Biological Triples by Θ_Σ

Node 1	Node 2	Regulatory Node	CMI	Θ_Σ	Θ_T	Σ	P_Σ	P_T	S
HOXA3	*HOXA5*	*PBX3*	0.820316	8.265877	8.048662	0.437625	0.001	0.001	0.511452
HOXA5	*HOXA3*	*LMX1B*	0.973260	7.506890	6.763280	0.401504	0.001	0.001	0.631465
HOXB5	*HOXB6*	*MEIS1*	0.819451	7.205339	6.846796	0.427434	0.001	0.001	0.501928
HOXA2	*HOXA5*	*MEIS1*	0.573735	7.019063	5.996416	0.367280	0.001	0.001	0.596266
HOXA2	*HOXA5*	*MEF2A*	0.704308	6.448953	6.374430	0.357321	0.001	0.001	0.618474
HOXA3	*HOXA2*	*MEF2A*	0.736876	6.229782	5.449352	0.357688	0.001	0.001	0.618665
HOXA9	*HOXA3*	*MEIS1*	0.612700	6.147447	5.840250	0.376088	0.001	0.001	0.516719
HOXA1	*HOXB7*	*DLX3*	0.319020	6.031555	5.996593	0.377072	0.001	0.001	0.719936
HOXA3	*HOXA7*	*LMX1B*	0.796522	5.930516	5.361882	0.315926	0.001	0.001	0.575184
HOXA9	*HOXA7*	*LMX1B*	0.795180	5.921237	6.062873	0.351938	0.001	0.001	0.520730
HOXB3	*HOXB6*	*HOXB7*	0.382231	5.755689	4.911239	0.320005	0.001	0.001	0.636879
HOXB6	*HOXB3*	*MAX*	0.711938	5.676961	6.039347	0.309040	0.001	0.001	0.535384
HOXA3	*HOXA2*	*MEIS1*	0.632230	5.643140	5.226696	0.343473	0.001	0.001	0.650853
HOXB1	*PBX3*	*DMRTC2*	0.200534	5.582683	5.388206	0.305488	0.001	0.001	0.566194
HOXB5	*HOXB3*	*HOXA7*	0.504192	5.464608	5.008018	0.320871	0.001	0.001	0.519581
HOXB5	*HOXB6*	*HOXA3*	0.719784	5.333075	5.807730	0.357958	0.001	0.001	0.5175458
HOXA2	*HOXA5*	*LMX1B*	0.685202	5.254939	5.041743	0.314368	0.001	0.001	0.710364
HOXA3	*HOXA2*	*HOXB3*	0.614658	5.212560	5.482464	0.325482	0.001	0.001	0.669220
HOXB2	*HOXB3*	*SPI1*	0.992869	5.194840	5.064013	0.338784	0.001	0.001	0.542340
HOXA1	*HOXB8*	*RUNX3*	0.176414	5.148190	4.848085	0.286748	0.001	0.001	0.626724
HOXA5	*HOXA2*	*SIX3*	0.697338	5.129497	5.577772	0.309603	0.001	0.001	0.542347
HOXB7	*HOXB6*	*MLXIPL*	0.511277	5.113759	5.390683	0.361182	0.001	0.001	0.542626
HOXA7	*PBX3*	*POU4F1*	0.555084	5.111917	5.837126	0.312872	0.001	0.001	0.586764
HOXB8	*HOXA1*	*DLX3*	0.206524	5.058267	5.102255	0.278793	0.001	0.001	0.609568
HOXB6	*HOXB7*	*STAT5B*	0.563708	5.048295	4.784504	0.355652	0.001	0.001	0.569296
HOXA3	*HOXA2*	*HOXB8*	0.704156	4.982283	4.596447	0.326155	0.001	0.001	0.5670250
HOXB1	*PBX3*	*SREBF1*	0.238224	4.964400	3.933486	0.290385	0.001	0.001	0.5993946
MEIS1	*HOXA3*	*NR6A1*	0.484028	4.787769	4.564621	0.384765	0.001	0.001	0.5612223
HOXA3	*HOXA2*	*ERG*	0.854079	4.775633	4.719983	0.307792	0.001	0.001	0.5308264
HOXB3	*HOXB5*	*IRX1*	0.681236	4.762596	4.095362	0.287235	0.001	0.001	0.5965832
HOXB2	*HOXB6*	*HOXB7*	0.349210	4.660569	3.497816	0.316489	0.001	0.002	0.6341667
HOXB5	*HOXB6*	*HOXA9*	0.728788	4.625061	3.950498	0.325499	0.001	0.001	0.5515168
HOXA9	*HOXB7*	*MLXIPL*	0.168949	4.618047	3.982716	0.289461	0.001	0.001	0.6315528
HOXB3	*HOXB5*	*HOXA5*	0.547398	4.574438	5.077232	0.286324	0.001	0.001	0.6519540
HOXB1	*HOXA5*	*MAFA*	0.258676	4.553432	4.648329	0.279503	0.001	0.001	0.6090695
HOXA1	*HOXB6*	*ATF4*	0.371474	4.544329	4.315940	0.299968	0.001	0.001	0.6979978
MEIS1	*HOXA5*	*HEYL*	0.488968	4.526637	3.193835	0.382446	0.001	0.002	0.6221348
HOXB2	*HOXB5*	*IRX1*	0.661687	4.473764	4.903444	0.295175	0.001	0.001	0.6057017
HOXB7	*HOXB6*	*HOXA9*	0.346855	4.444519	3.487488	0.328127	0.001	0.002	0.5408434
HOXB3	*HOXB7*	*CEBPA*	0.456098	4.437850	5.265165	0.302595	0.001	0.001	0.7035531
HOXB8	*HOXA1*	*PLAGL1*	0.174812	4.435822	4.227541	0.259337	0.001	0.001	0.6397053
PBX3	*HOXA3*	*DLX4*	0.376259	4.341928	3.383559	0.294993	0.001	0.003	0.5599292
HOXA1	*HOXA5*	*RARA*	0.564074	4.283333	4.080688	0.347871	0.001	0.002	0.7039146
HOXB7	*HOXB5*	*CEBPA*	0.523503	4.248256	4.572174	0.316662	0.001	0.001	0.5829643
HOXB8	*HOXA9*	*IRX5*	0.143996	4.244078	3.743148	0.255930	0.001	0.002	0.5426364
HOXA7	*PBX3*	*MEIS2*	0.481629	4.229047	4.704656	0.291258	0.001	0.001	0.5720851
HOXA1	*HOXA3*	*JDP2*	0.462743	4.227947	4.378695	0.349784	0.001	0.001	0.6529631
MEIS1	*PBX3*	*FOXI1*	0.399453	4.217408	3.765785	0.351739	0.001	0.001	0.5882254
HOXB7	*HOXB6*	*HOXA2*	0.396262	4.213586	3.643030	0.323173	0.001	0.001	0.6443882
HOXA2	*HOXB7*	*PBX1*	0.297987	4.206895	2.752253	0.303415	0.001	0.005	0.6995975

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TABLE S3: Triples with the 50 highest Θ_Σ-values. Chosen from the analysed triples where only the five triples with highest Θ_Σ-values for each edge are considered and after the removal of 50 outlier genes.

Node 1	Node 2	Reg. Node	CMI	Θ_Σ	Θ_T	Σ	P_Σ	P_T	S
GFI1B	*HMG20B*	*MAFG*	0.159214	6.728150	6.919335	0.239888	0.001	0.001	0.656489
ESR1	*ESRRA*	*CUX1*	0.136866	6.633394	6.387613	0.257335	0.001	0.001	0.740265
ATF2	*CREB3L2*	*NKX2-3*	0.183120	6.597125	6.372863	0.292662	0.001	0.001	0.699403
AR	*ERG*	*DLX3*	0.181392	6.419430	6.294301	0.264732	0.001	0.001	0.670266
GMEB1	*GMEB2*	*GABPA*	0.160774	6.393412	6.419077	0.249189	0.001	0.001	0.648156
TP53	*TOPORS*	*RFX1*	0.132448	6.245761	6.067981	0.245629	0.001	0.001	0.636249
SIX3	*NR4A3*	*PPARD*	0.126805	6.157518	5.990863	0.226794	0.001	0.001	0.626979
CREM	*CREB3*	*GABPA*	0.145051	6.013678	6.087598	0.217777	0.001	0.001	0.705945
NR1H2	*RXRB*	*FOXO1*	0.210384	5.997208	5.846367	0.283394	0.001	0.001	0.622814
AIRE	*GMEB1*	*ZSCAN16*	0.139642	5.986620	5.997924	0.230049	0.001	0.001	0.742193
MAFG	*NFE2*	*SMARCC1*	0.133617	5.963117	5.929599	0.233347	0.001	0.001	0.675628
MAFG	*NFE2*	*MBD2*	0.129252	5.962294	5.873812	0.233020	0.001	0.001	0.656413
EGR2	*HOXB2*	*HOXA9*	0.156973	5.865105	5.867066	0.228347	0.001	0.001	0.688156
MEIS1	*PBX1*	*ZEB1*	0.188129	5.842200	5.931817	0.265427	0.001	0.001	0.611296
WT1	*SOX12*	*ARX*	0.172959	5.825481	5.423547	0.310742	0.001	0.001	0.622995
MYOD1	*TCF12*	*RFX1*	0.134786	5.809604	5.792604	0.202536	0.001	0.001	0.574806
KDM2B	*ZNF784*	*TEAD1*	0.184460	5.808177	6.087558	0.237961	0.001	0.001	0.699574
ELF1	*TFDP1*	*KDM2B*	0.180421	5.783468	6.248786	0.216104	0.001	0.001	0.665745
ZNF423	*EBF1*	*ZIC4*	0.134866	5.766551	5.550101	0.245472	0.001	0.001	0.714254
TP53	*IRF5*	*BCL6*	0.185771	5.751306	6.407961	0.206465	0.001	0.001	0.612062
ESR1	*XBP1*	*HEY2*	0.217111	5.741680	5.731264	0.262501	0.001	0.001	0.707043
MEIS3	*PBX3*	*DRGX*	0.116614	5.726887	5.518357	0.225603	0.001	0.001	0.640716
FOS	*ELK1*	*RORA*	0.128186	5.714594	5.384268	0.256372	0.001	0.001	0.698619
LMX1B	*GBX1*	*MSX2*	0.114587	5.677847	5.501699	0.227926	0.001	0.001	0.626511
CEBPA	*CEBPB*	*SMARCC1*	0.149690	5.675830	5.864768	0.232468	0.001	0.001	0.629392
AR	*HOXB13*	*IRX5*	0.127068	5.628973	5.607486	0.227873	0.001	0.001	0.677219
LEF1	*CDX1*	*CEBPG*	0.173025	5.582373	5.740356	0.204470	0.001	0.001	0.589337
JUN	*ATF2*	*IRF6*	0.160900	5.540512	5.706251	0.217531	0.001	0.001	0.743458
LMX1B	*FEV*	*CEBPE*	0.132053	5.459481	5.381255	0.217621	0.001	0.001	0.531703
DLX6	*HAND2*	*GFI1*	0.316684	5.447074	5.452548	0.409997	0.001	0.001	0.5666130
EGR3	*NFATC2*	*FOSB*	0.139154	5.444294	5.407629	0.230926	0.001	0.001	0.7545026
ARNT2	*EPAS1*	*STAT5B*	0.184543	5.421202	5.515499	0.269106	0.001	0.001	0.6476382
HOMEZ	*HMBOX1*	*POU3F2*	0.116605	5.417860	5.441199	0.199304	0.001	0.001	0.6313812
AR	*RREB1*	*LMO2*	0.114581	5.415712	5.247730	0.221936	0.001	0.001	0.7792968
NPAS2	*RORA*	*TOPORS*	0.139477	5.395178	5.542092	0.214479	0.001	0.001	0.7393471
FOXP2	*SP4*	*TBX3*	0.139812	5.377868	5.488782	0.217301	0.001	0.001	0.5843965
GATA1	*IKZF1*	*DPRX*	0.130265	5.357680	5.120758	0.233326	0.001	0.001	0.7042830
NANOG	*TET1*	*PURA*	0.142164	5.348978	5.556867	0.212450	0.001	0.001	0.7522146
KDM2B	*ZNF740*	*NOBOX*	0.103249	5.343459	5.074403	0.205580	0.001	0.001	0.7296110
PITX2	*LHX3*	*TBX5*	0.156647	5.315290	4.504254	0.193797	0.001	0.001	0.6662439
AR	*ERG*	*DDIT3*	0.138062	5.307119	5.147755	0.236228	0.001	0.001	0.6881829
KDM2B	*ZNF232*	*MZF1*	0.102529	5.306776	5.161613	0.197083	0.001	0.001	0.6727312
ARID2	*SMARCC2*	*VDR*	0.106636	5.273111	5.122535	0.203007	0.001	0.001	0.6993566
ESR1	*NR2C2*	*HOXD8*	0.129780	5.247737	5.008668	0.234352	0.001	0.001	0.7039140
LMO2	*TAL1*	*ZNF32*	0.105264	5.224616	5.238358	0.182179	0.001	0.001	0.7148965
ARNT	*EPAS1*	*REST*	0.121260	5.167161	5.191644	0.214579	0.001	0.001	0.7482859
LMO2	*NHLH1*	*TBX19*	0.102305	5.166281	5.103774	0.196432	0.001	0.001	0.7178833
E2F3	*TFDP1*	*BPTF*	0.184846	5.118170	4.863077	0.203638	0.001	0.001	0.6571473
SMAD4	*SMAD5*	*HOXB6*	0.443231	5.113130	5.490316	0.374981	0.001	0.001	0.6213580
MITF	*PAX3*	*MLXIPL*	0.131645	5.102654	4.907555	0.173291	0.001	0.001	0.6373616

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TABLE S4: Genes included in the 50 triples with highest Θ_Σ-value. The list includes genes relevant to AML found in literature (left) and a list of triples (right). Genes with relevance to AML have been bolded.

Gene	Reference	Gene 1	Gene 2	Regulatory Gene
ARID2	[10] T. Bluemn, et al. (2022)	***GFI1B***	*HMG20B*	*MAFG*
ARNT	[11] S. Williams, et al. (2010)	***ESR1***	*ESRRA*	***CUX1***
BCL6	[12] K. Kawabata, et al. (2021)	*ATF2*	*CREB3L2*	***NKX2-3***
BPTF	[13] A. Radzisheuskaya, et al. (2023)	*AR*	***ERG***	*DLX3*
CEBPA	[14] A. Fasan, et al. (2014)	*GMEB1*	*GMEB2*	*GABPA*
CEBPB	[15] W. Fu, et al. (2022)	***TP53***	*TOPORS*	*RFX1*
CEBPE	[16] Li, K. et al. (2019)	*SIX3*	***NR4A3***	*PPARD*
CEBPG	[17] Y. Jiang, et al. (2021)	*CREM*	***CREB3***	*GABPA*
CREB3	[18] S. Feng, et al. (2020)	*NR1H2*	*RXRB*	***FOXO1***
CUX1	[19] M. E. McNerney, et al. (2013)	*AIRE*	*GMEB1*	*ZSCAN16*
ELF1	[20] P. Varghese, et al. (2025)	*MAFG*	***NFE2***	*SMARCC1*
ELK1	[21] D. Guo, et al. (2023)	*MAFG*	***NFE2***	***MBD2***
EPAS1	[22] S. Wang, et al. (2023)	*EGR2*	***HOXB2***	***HOXA9***
ERG	[23] G. Marcucci, et al. (2005)	***MEIS1***	*PBX1*	***ZEB1***
ESR1	[24] A. Roma, et al. (2020)	***WT1***	***SOX12***	*ARX*
FEV	[25] J. Zhang, et al. (2022)	***MYOD1***	*TCF12*	*RFX1*
FOS	[26] F. Yang, et al. (2024)	***KDM2B***	*ZNF784*	*TEAD1*
FOSB	[27] SH. Luan, et al. (2022)	***ELF1***	*TFDP1*	***KDM2B***
FOXO1	[28] S. Lin, et al. (2014)	*ZNF423*	*EBF1*	*ZIC4*
GATA1	[29] R. Ayala, et al. (2009)	***TP53***	*IRF5*	***BCL6***
GFI1	[30] T. Möröy, et al. (2015)	***ESR1***	*XBP1*	*HEY2*
HOXA9	[31] S. Aryal, et al. (2023)	*MEIS3*	***PBX3***	*DRGX*
HOXB13	[32] Y. Chu, et al. (2018)	***FOS***	***ELK1***	***RORA***
HOXB2	[33] O. Lindblad, et al. (2015)	*LMX1B*	*GBX1*	*MSX2*
HOXB6	[34] A. Giampaolo, et al. (2002)	***CEBPA***	***CEBPB***	*SMARCC1*
IKZF1	[35] J. Eckardt, et al. (2023)	*AR*	***HOXB13***	***IRX5***
IRX5	[36] S. Nagel, et al. (2022)	***LEF1***	*CDX1*	***CEBPG***
JUN	[37] C. Zhou, et al. (2017)	***JUN***	*ATF2*	*IRF6*
KDM2B	[38] V. van den Boom, et al. (2016)	*LMX1B*	*FEV*	***CEBPE***
LEF1	[39] K. Feder, et al. (2020)	*DLX6*	*HAND2*	***GFI1***
LMO2	[40] L. Lu, et al. 2023	*EGR3*	***NFATC2***	***FOSB***
MBD2	[41] K. Zhou, et al. (2021)	*ARNT2*	***EPAS1***	***STAT5B***
MEIS1	[42] U. Thorsteinsdottir, et al. (2001)	*HOMEZ*	*HMBOX1*	*POU3F2*
MYOD1	[43] M. Toyota, et al. (2001)	*AR*	*RREB1*	***LMO2***
NFATC2	[44] S. D. Patterson, et al. (2021)	***NPAS2***	***RORA***	*TOPORS*
NFE2	[45] J. S. Jutzi, et al. (2019)	*FOXP2*	*SP4*	*TBX3*
NKX2-3	[46] S. Nagel, et al. (2021)	***GATA1***	***IKZF1***	*DPRX*
NPAS2	[47] B. Song, et al. (2018)	*NANOG*	***TET1***	***PURA***
NR4A3	[48] SC. Lin, et al. (2022)	***KDM2B***	*ZNF740*	*NOBOX*
PBX3	[49] G. J. Dickson, et al. (2013)	*PITX2*	*LHX3*	*TBX5*
PURA	[50] K. Lezon-Geyda, et al. (2001)	*AR*	***ERG***	*DDIT3*
RORA	[51] C. Snider, et al. (2019)	***KDM2B***	*ZNF232*	*MZF1*
SMAD4	[52] Y. Imai, et al. (2001)	***ARID2***	*SMARCC2*	*VDR*
SOX12	[53] H. Wan, et al. (2017)	***ESR1***	*NR2C2*	*HOXD8*
STAT5B	[54] B. Maurer, et al. (2019)	***LMO2***	***TAL1***	*ZNF32*
TAL1	[22] Z. Wang, et al. (2023)	***ARNT***	***EPAS1***	*REST*
TET1	[55] J. Wang, et al. (2018)	***LMO2***	*NHLH1*	*TBX19*
TP53	[56] K. Barbosa, et al. (2019)	*E2F3*	*TFDP1*	***BPTF***
WT1	[57] R. Rampal, et al. (2016)	***SMAD4***	*SMAD5*	***HOXB6***
ZEB1	[58] W. G. Shousha, et al. (2019)	*MITF*	*PAX3*	*MLXIPL*

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